

Mahmoud Alyosify

Software Engineer | .NET & Python Backend, Data Systems & AI-Enabled Applications | MSc Artificial Intelligence, Queen's University (Canada)

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🌐 [linkedin.com/in/mahmoudalyosify](https://www.linkedin.com/in/mahmoudalyosify) | 🐙 github.com/MahmoudAlyosify | 📁 Portfolio | 🤖 Hugging Face | 🏆 Kaggle

PROFESSIONAL SUMMARY

Software Engineer with production experience on **national-scale infrastructure** — backend systems for Egyptian toll operations serving **50%+ of national vehicles** — built in **C#, .NET, Entity Framework and SQL Server**. Full-stack trained (**Route Academy .NET & Angular diploma**, plus **300+ hours** of ITI development courses in .NET, SQL Server, MEARN stack and Android), and equally comfortable shipping **Python services with FastAPI, Docker and AWS**. Now completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship** — bringing the increasingly rare combination of **solid software engineering fundamentals** and **genuine AI/ML depth** to teams building AI-enabled products.

EDUCATION

- MSc in Artificial Intelligence**
Queen's University, Kingston, Ontario, Canada

Sep 2025 – Nov 2026
Expected Nov 2026

 - **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
 - Coursework: **Cloud Computing, End-to-End ML Pipelines, Large Language Models, Deep Learning, Reinforcement Learning.**
- B.Sc. Computer & Information Science (Bioinformatics)**
Assiut University, Egypt

Sep 2019 – Jul 2023
GPA: 3.53 / 4.00

 - Foundations in data structures, algorithms, operating systems, databases and discrete mathematics; approx. **100 competitive-programming problems** solved on Codeforces.
 - Graduation Project: **Vitalism Solution** — **Grade A+; Semi-Finalist, Microsoft Imagine Cup 2023.**

TECHNICAL SKILLS

Backend & .NET	C#, .NET, ASP.NET Core, ASP.NET MVC, Web API, Entity Framework, ADO.NET, LINQ, WinForms, Crystal Reports, Stored Procedures, Triggers, RESTful API Design
Python & Services	Python, FastAPI, Flask, Streamlit, Docker, AWS, CI/CD, ONNX, Background Jobs & Pipelines
Frontend	Angular, TypeScript, JavaScript, HTML5, CSS, Bootstrap, React (MEARN training)
Databases	SQL Server, MySQL, MongoDB, Database Design & Normalisation, Query Optimisation, ETL Pipelines
AI Integration	LLM APIs, Prompt Engineering, RAG, Fine-Tuning (LoRA, QLoRA), PyTorch, TensorFlow, Scikit-learn, FAISS, ChromaDB, Agentic Coding Tools (Claude Code, Cursor, Copilot)
Languages & Tools	C#, Python, C++, C, JavaScript, TypeScript, SQL, Go, Java, R, MATLAB, Git/GitHub, Linux, Bash

PROFESSIONAL EXPERIENCE

- Systems Engineer (Military Service)**
National Company for Roads Building & Development

Feb 2024 – Mar 2025
Cairo, Egypt

 - Supported and maintained backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles** across **70%+ of national roads**, working within the Control and Command environment.
 - Co-developed an internal multi-department **data-management and reporting desktop application** using **C#, .NET, Entity Framework, SQL Server** and **Crystal Reports**, replacing manual reporting across stores and departments.
 - Debugged and optimised **real-time transportation software** and provided technical support for legacy applications, minimising downtime and ensuring operational continuity.
- Machine Learning Instructor**
National Telecommunication Institute (NTI)

Jul 2025 – Sep 2025
Cairo, Egypt

 - Delivered a **90-hour** hands-on ML program for engineering and CS students, guiding them through building complete solutions on real datasets.
- AI in Cybersecurity Intern**
e& Egypt — via NTI (On-the-Job Training)

Jun 2025 – Jul 2025
Cairo, Egypt

 - Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.
- Online Instructor — Programming, Algorithms & Data Science**
Udemy & YouTube

2020 – Present
Remote

 - Authored and taught Arabic-language courses in discrete mathematics, algorithms, probability, statistics and graph data mining for **10,000+ learners (700K+ views)**.

SELECTED SOFTWARE PROJECTS

Inventory Management System

C# · ASP.NET Core · SQL Server · Barcode Scanning

- Built a CRUD-based inventory system with barcode scanning and alert notifications, covering database schema design through user-facing workflow.

E-Commerce Web Application

Angular · Bootstrap · Stripe API · JWT Authentication

- Developed a frontend shopping experience with secure JWT-based login and **Stripe** payment integration.

Mini-FAT File System

C · Systems Programming · Operating Systems

- Implemented a FAT-style file system in C from first principles — allocation tables, directory structures and file operations.

HORUS Sentinel — Multi-Agent Intelligence Platform

2026

Python · Multi-Agent Architecture · Knowledge Graphs · Self-Hosted LLM

- Architected a distributed system of specialised autonomous agents writing into a shared knowledge-graph state layer, with a self-hosted model generating evidence-backed reports.

Additional Projects

- **Horus-OSINT** — fine-tuned **Meta-Llama-3-8B** on **159,826** records and deployed it on **AWS** as an interactive service.
- **AudioShield** — CNN-based audio classifier served through a Streamlit application with automated PDF reporting.
- **Human vs. Agentic Pull Requests** — reproducible GitHub-mining pipeline analysing code-review effort across large open-source repositories.
- **Also:** Student Affairs System · Vitalism Solution (rPPG) · SCRAP · SimCLR-Vision-SSL (ONNX/FAISS) · ATM Transaction Analysis.

CERTIFICATIONS & AWARDS

Full Stack Development Diploma — .NET & Angular (Route Academy) **.NET Web Development** — ITI (120 hrs)

SQL Server & C# — ITI (90 hrs)

MEARN Stack Development — ITI (60 hrs)

AWS Academy — ML Foundations, ML for NLP, Data Engineering **Microsoft Certified: Data Analyst Associate**

1st Place — Smart Cities Hackathon 2022

Semi-Finalist — Microsoft Imagine Cup 2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency · **French:** Basic.